

Arduino-Based Electronic Voting Machine

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Abstract

An Electronic Voting Machine (EVM) is an electronic system designed to record votes as an alternative to traditional ballot-based voting. This project presents a basic electronic voting machine developed using an Arduino UNO and associated electronic components. The system records votes through push buttons, increments the corresponding vote count, and displays voting information on a 16×2 LCD. A control mechanism is included to manage voting inputs, display results, and reset the system. The project demonstrates a simple and low-cost approach to electronic vote recording and result display for educational applications.

Introduction

This project focuses on the development of a basic Electronic Voting Machine (EVM) using an Arduino UNO and associated electronic components. The system operates by increasing the corresponding vote count when a voting input button is pressed. Once a vote is registered, the voting inputs are disabled until the MASTER button is manually set to HIGH, providing a basic control mechanism to prevent repeated inputs during the voting process.

The number of participating parties can be selected using the ENABLE input switches. A RESET button is provided to return the system to its initial state. When the RESULT switch is set to HIGH, the corresponding vote counts are displayed on the LCD, allowing the winning candidate to be determined based on the recorded votes.

Figure 1: Circuit Diagram of the Arduino-Based Electronic Voting Machine

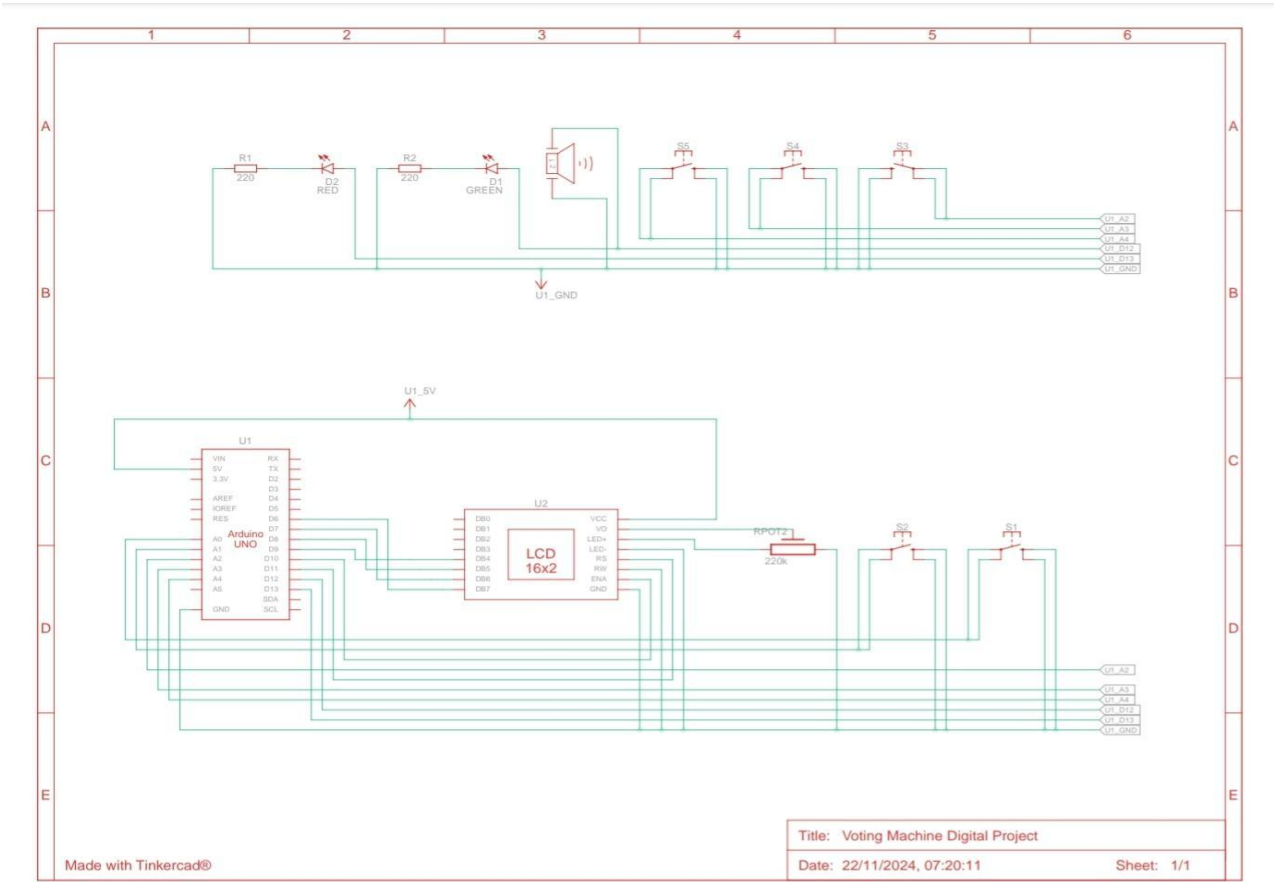
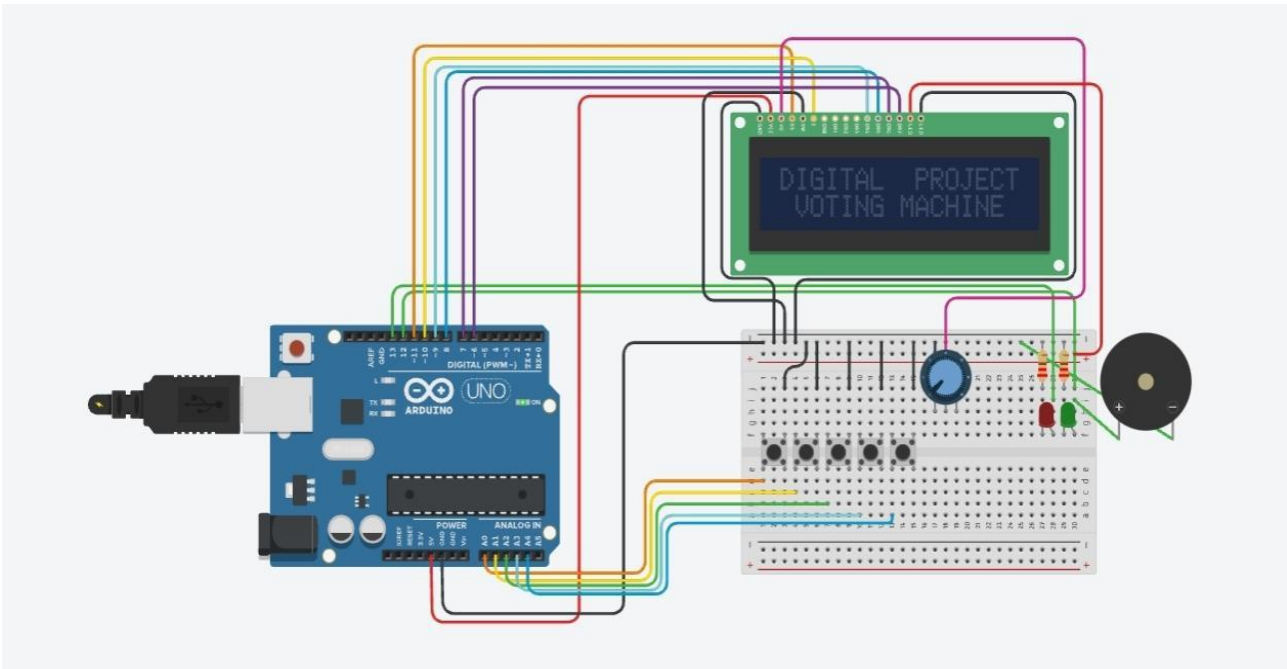


Figure 2: Simulation of the Arduino-Based Electronic Voting Machine



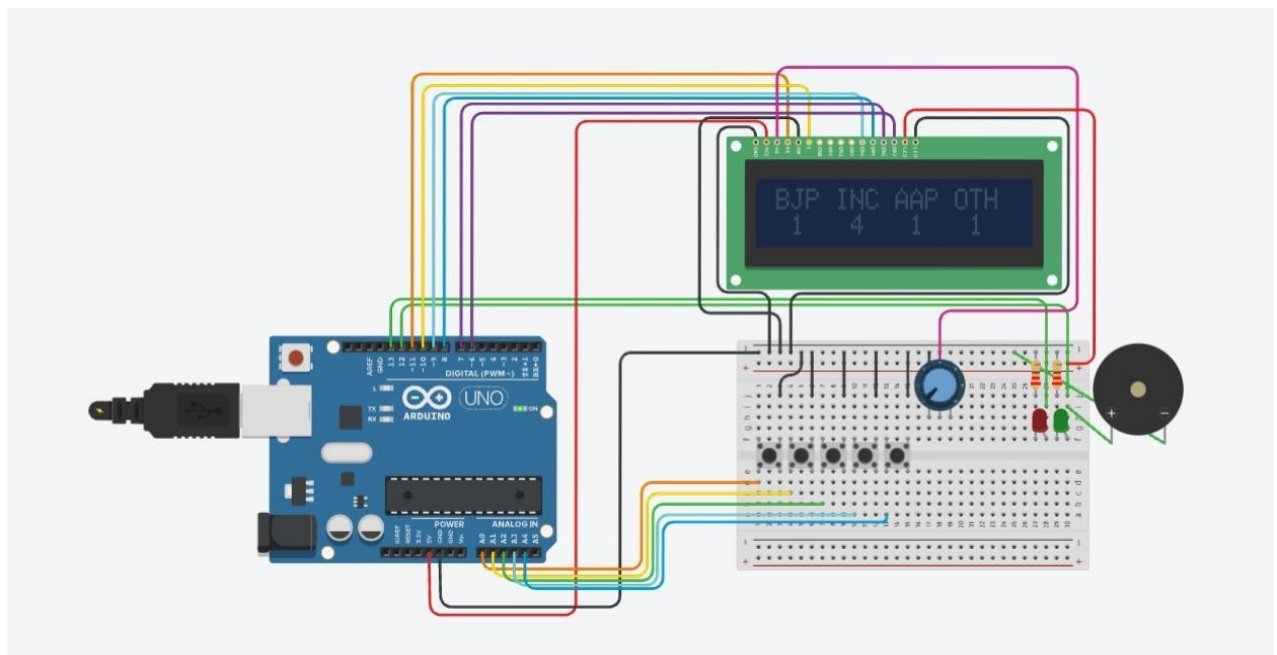
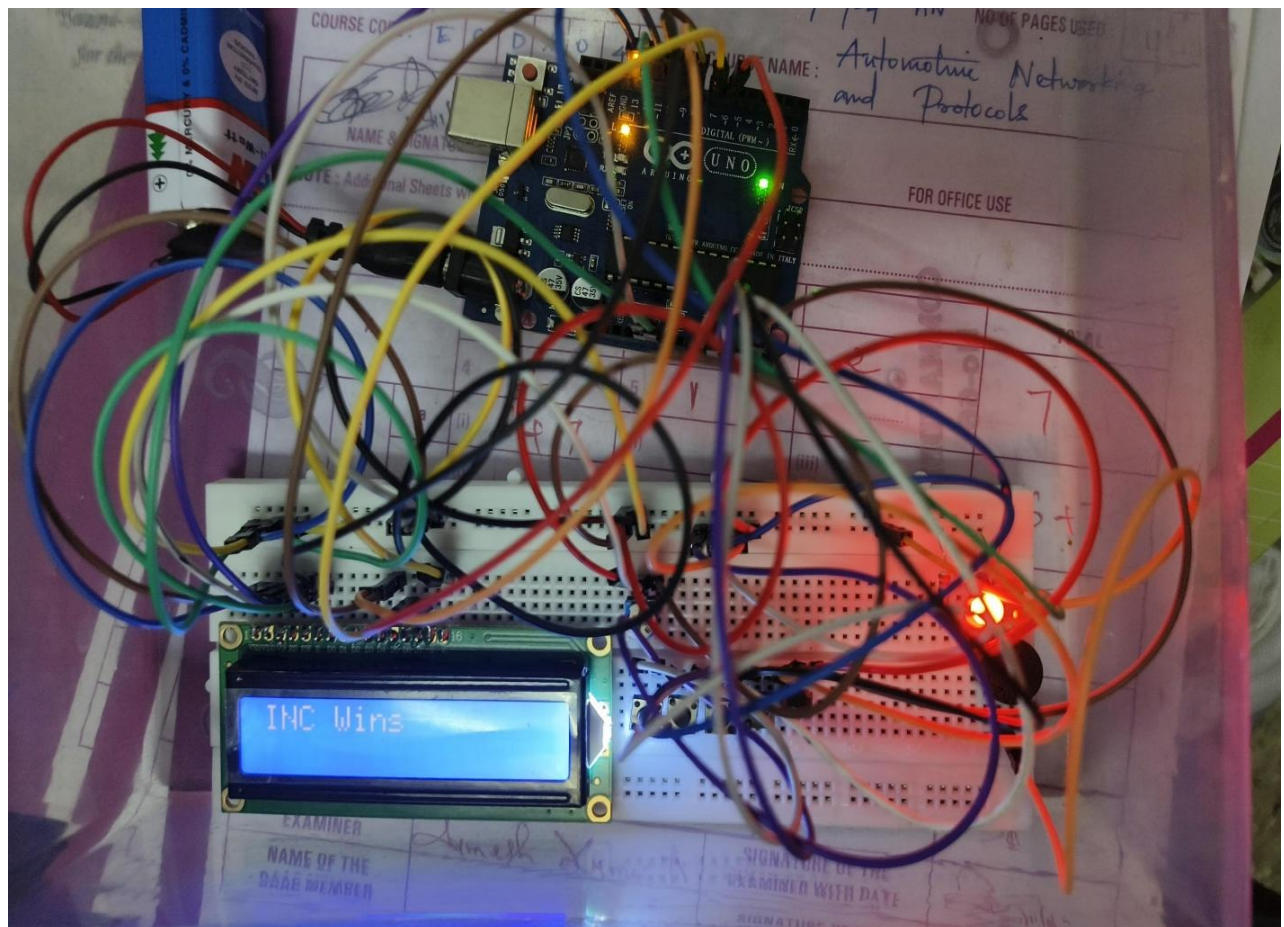


Figure 3: Hardware Prototype of the Arduino-Based Electronic Voting Machine



Components Required

1. Arduino UNO

The Arduino UNO is the main microcontroller used in the Electronic Voting Machine. It is based on the ATmega328P and provides 14 digital input/output pins, 6 analog inputs, and a 16 MHz clock. In this project, the Arduino UNO receives inputs from the push buttons, processes the voting information, controls the LCD display, and manages the LED and buzzer outputs. It also handles the vote-counting and result-display operations of the system.

2. 16×2 LCD

A 16×2 Liquid Crystal Display (LCD) is used to display information in the Electronic Voting Machine. It can display 16 characters in each of its two rows. In this project, the LCD is interfaced with the Arduino UNO to display voting information, vote counts, and the final result. It provides clear visual feedback to the user during the operation of the voting system.

3. Potentiometer

A 10k potentiometer is used with the 16×2 LCD to adjust its contrast. It helps make the characters displayed on the LCD clearly visible.

4. Push Buttons

Push buttons are used as the voting inputs in the Electronic Voting Machine. Each button represents a voting option, and pressing a button sends an input signal to the Arduino UNO. The Arduino detects the button press and increments the corresponding vote count.

5. Resistors

Resistors are used in the Electronic Voting Machine to limit current and provide appropriate electrical conditions for the connected components. In this project, resistors are used with the LEDs to control the current flowing through them and protect the LEDs from excessive current. They help ensure the reliable operation of the electronic components in the voting circuit.

6. LEDs

LEDs are used as visual indicators in the Electronic Voting Machine. They provide an indication of the system status during operation and help the user identify the corresponding circuit activity. The LEDs are connected to the Arduino-controlled circuit and operate according to the programmed system conditions.

7. Jumper Wires

Jumper wires are used to establish electrical connections between the Arduino UNO, LCD, push buttons, LEDs, buzzer, and other components of the Electronic Voting Machine. They provide the necessary connections for transferring power and signals between the components and allow the complete circuit to operate as intended.

8. Buzzer

The buzzer is used to provide audible feedback during the operation of the Electronic Voting Machine. In this project, the buzzer produces a sound when a voting input is registered, providing confirmation to the user that the button press has been detected. It is controlled by the Arduino according to the programmed voting operation.

Circuit Operation

The operation of the Electronic Voting Machine consists of the following main steps:

1. Initialization

When the system is powered on, the Arduino UNO initializes the 16×2 LCD and sets the required input and output pins. The voting buttons are configured as inputs, while the LEDs and buzzer are configured as outputs. The vote-count variables are initialized to zero, preparing the system for the voting process.

2. Button Press Detection

After initialization, the Arduino continuously checks the voting buttons for user input. Each button represents a voting option and is connected to an input pin of the Arduino UNO. When a button is pressed, the Arduino detects the corresponding input signal and registers the selected voting option.

3. Vote Counting

When a valid voting button is pressed, the Arduino increments the corresponding vote-count variable. The buzzer produces an audible indication to confirm that the vote has been registered. The LCD provides the required voting information during the operation.

4. Result Display

When the result control is activated, the Arduino compares the recorded vote counts to determine the candidate with the highest number of votes. The corresponding result and winning candidate are then displayed on the 16×2 LCD.

5. Reset

After the result is displayed, the system can be returned to its initial state using the reset operation. The vote counts are cleared, the LCD is reset, and the system is prepared for a new voting session.

Result and Discussion

The developed Electronic Voting Machine successfully accepts voting inputs through push buttons and processes the corresponding vote counts using the Arduino UNO. The 16×2 LCD provides visual feedback during the voting process and displays the recorded results when the result control is activated. The buzzer provides audible feedback when a vote is registered. The prototype demonstrates the basic operation of an Arduino-based electronic voting system.

Conclusion

The Arduino-Based Electronic Voting Machine was successfully developed as a basic prototype for electronic vote recording and result display. The system uses push buttons for voting input, an Arduino UNO for processing, and a 16×2 LCD for displaying information and results. The project demonstrates the practical use of embedded systems concepts such as digital input handling, vote counting, display interfacing, and system control.

Reference

This project was developed as an academic project with reference to an online tutorial. The tutorial was studied to understand the circuit implementation and programming approach used for the Electronic Voting Machine.

Reference Video

<https://www.youtube.com/watch?v=yeEX9Z7e64I>